

Mining Congestion Propagation Patterns in Urban Road Networks: A Reinforcement Learning Method

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Abstract—Congestion propagates in urban road networks following latent patterns, constrained by the physical topology of the road network and influenced by human mobility behavior. Exploring and leveraging these patterns can provide decision-making support for traffic management authorities. However, existing methods typically assume independent congestion effects, neglecting the sequential impacts of congestion propagation and failing to capture long-term dependencies in the spreading process. To address these limitations, we propose a novel reinforcement learning-based congestion propagation pattern mining method (named GRL_CPP). This approach formalizes congestion propagation as a Markov Decision Process (MDP) and employs reinforcement learning algorithms to continuously optimize decisions, automatically learning long-term dependencies within congestion propagation sequences. With the exploration mechanism of reinforcement learning, GRL_CPP can actively identify the most discriminative propagation patterns in complex traffic environments while avoiding unrealistic assumptions. Extensive experiments on both real-world and synthetic datasets demonstrate that our method significantly outperforms state-of-the-art baselines, achieving over 41% average improvements in real-world bidirectional congestion analysis across various data sparsity scenarios, and 74% improvement in the F1-score for true propagation graph reconstruction.

Index Terms—Traffic congestion, spatiotemporal data analysis, propagation pattern, graph reinforcement learning.

I. INTRODUCTION

INFLUENCED by the physical topology of road networks and the dynamics of human behavior, congestion propagation in urban road networks follows latent patterns. Identifying and leveraging these patterns can help locate bottlenecks in the road network [1], [2], [3], predict future congestion conditions [4], [5], and guide urban road planning [6], thereby enhancing the overall efficiency and capacity of transportation systems. Over the past few decades, extensive research has been devoted to analyzing latent congestion propagation patterns in road networks. Early studies often simplified the problem by partitioning networks into uniform grids [7], [8], which inherently obscured fine-grained topological details and reduced model sensitivity

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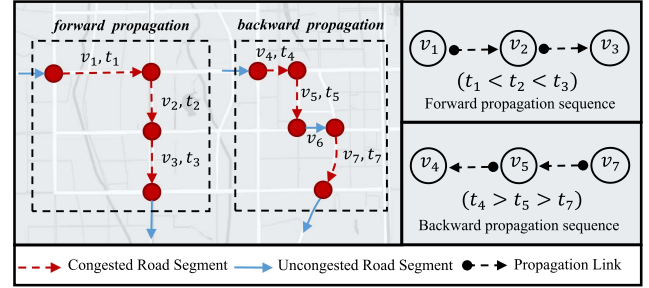


Fig. 1. Congestion propagation in the road network.

to local congestion dynamics. Alternatively, simulation-based approaches utilized traffic flow theory [9] and epidemiological models [10] to deduce propagation from simulated data or limited real-world observations. However, these analytical approaches heavily rely on underlying models and idealized assumptions, limiting their generalization to asymmetric real-world traffic environments.

In recent years, the increasing number of vehicle-mounted positioning devices and growing demand for navigation services have led to an explosion of traffic data, providing strong support for solving problems in traffic congestion detection [11] and prediction [12], bottleneck identification [1], [2], [3], and traffic signal control [13], [14], [15]. Leveraging these large-scale, high-precision traffic datasets, researchers develop statistical-based methods [16], [17], [18] to mine congestion propagation patterns in urban road networks. However, these methods assume that congestion propagation are mutually independent, ignoring the long-term and sequential dependencies inherent in the propagation process. Addressing these limitations presents the following challenges:

- 1) *Sequentiality of congestion propagation*: The physical topology of the road network constrains congestion to propagate only along or against the direction of roads, thereby forming congestion propagation sequences. Within these sequences, each road segment is influenced by its precursors and transmits effects progressively. As displayed in Fig. 1, the congestion first appears on v_1 , then v_2 gets congested, and finally v_2 transmits effects to v_3 , causing congestion on v_3 .
- 2) *Heterogeneity of transportation infrastructure*: In an urban road network, the distinct attributes of roads and their varying external environments lead to diverse congestion dynamics. For example, v_6 is an arterial road with high

traffic capacity and numerous exit roads, allowing it to handle substantial traffic pressure. In contrast, v_5 is narrower and lacks sufficient exit neighbors, making it more prone to congestion.

- 3) *Complexity of spatiotemporal correlations:* The propagation of congestion across urban road networks exhibits intricate spatiotemporal correlations. Suppose we observe a congestion on v_5 , it may be directly affected by v_4 , or indirectly affected by v_7 , or even not affected by any other roads, thus we cannot assert which road should be responsible for the congestion on v_5 . Besides the spatial correlations, two road segments may experience congestion: (1) simultaneously, (2) sequentially within a small time interval, or (3) successively with a long time gap, which exhibits complex temporal correlations.

To address these challenges, we propose a novel reinforcement learning-based propagation pattern mining method, termed GRL_CPP. This approach models the long-sequence interaction of congestion propagation as a Markov Decision Process (MDP) and optimizes the decision process through a reinforcement learning (RL) algorithm. By leveraging the exploration mechanism of RL, the proposed method can proactively explore the most discriminative propagation patterns in the urban road network, avoiding unreasonable assumptions. Specifically, our main contributions are as follows:

- 1) We formulate the congestion propagation process as a MDP and optimize it using a RL algorithm. With the help of RL, our proposed method can effectively capture the sequential influences governing the propagation dynamics. To the best of our knowledge, we are the first to adopt a RL-based method to model the long-term sequential influence of congestion propagation and infer the congestion propagation patterns in road networks.
- 2) We embed a tailored Graph Convolutional Network (GCN) within the RL environment. This explicitly captures the heterogeneous infrastructural attributes of the road network and dynamic propagation states of the cascading path, providing the agent with highly contextualized state representations for accurate decisions.
- 3) We design a propagation constraint rule based on the spatial topology of the road network and the historical propagation sequence to ensure valid spatial propagation boundaries. More importantly, we formulate a novel reward function governed by time-lagged effects, enabling the agent to evaluate complex propagation correlations without relying on predefined thresholds.
- 4) Extensive experiments on two real-world datasets and two simulated datasets demonstrate that our proposed method effectively models long-sequence interaction behaviors in congestion propagation while maintaining better pattern mining capability compared to baselines, even in scenarios with sparse congestion data.

The rest of the article is organized as follows. Section II gives a brief literature review. Section III introduces symbols and definitions used in the article. Section IV details our proposed method. Section V presents the experimental setup and results.

Section VI concludes this work and provides future research directions.

II. RELATED WORK

A. Spatiotemporal Congestion Prediction

Research on spatiotemporal congestion prediction has progressed significantly, evolving from traditional probabilistic models to advanced deep learning architectures. Early studies predominantly relied on Markov chains, Bayesian networks, and historical spatial distribution matrices to infer congestion likelihood at localized road segments [7], [19], [20], [21]. Recently, deep learning-based spatiotemporal models have dominated the field. Researchers have extensively utilized Graph Convolutional Networks (GCNs) [22], [23], dynamic spatiotemporal Graph Neural Networks (GNNs) [24], attention mechanisms [25], and hybrid frameworks [26], [27] to automatically capture the dynamic spatiotemporal features of transportation networks. Furthermore, to address the sparsity and physical constraints inherent in traffic events, novel frameworks incorporating Physics-Informed Neural Networks (PINNs) [4], Neural Point Processes [5], and Ising models [28] have been proposed, achieving impressive predictive accuracy. However, while these approaches excel at predicting future traffic states, they focus primarily on the point-wise, localized evolutionary processes of congestion and fail to explicitly delineate the continuous, macroscopic trajectories of how congestion propagates across the network.

B. Congestion Propagation Pattern Analysis

To uncover the implicit diffusion mechanisms of traffic congestion, researchers have shifted focus towards discovering macroscopic propagation patterns. Several approaches draw analogies between traffic congestion and disease spread, employing models like SIS (Susceptible-Infected-Susceptible) and SIR (Susceptible-Infected-Recovered) to characterize diffusion and recovery rates of congestion [10], [29], [30]. Unfortunately, such analytical models heavily rely on strong idealized assumptions that often fail to reflect complex, dynamic traffic environments. Other studies employ frequent subgraph mining algorithms (e.g., STOTree [8], Apriori [17]) to extract spatiotemporal co-occurrence structures or subtrees from historical anomalies [6], [31], [32]. However, these methods intrinsically rely on predefined frequency thresholds and strict spatiotemporal continuity constraints, limiting their adaptability to heterogeneous real-world traffic environments. Another line of research models propagation as a network inference problem. For instance, [16] and [33] utilize pairwise classification or Dynamic Time Warping (DTW) to quantify segment correlations. Liang et al. [18] further formulated this as a submodular optimization problem to capture cascade patterns. Zeng et al. [34] designed a complex network-based approach to uncover spatiotemporal congestion patterns from traffic trajectories. Nevertheless, pairwise computations often ignore the multi-hop transitivity of congestion, and global optimization algorithms face severe

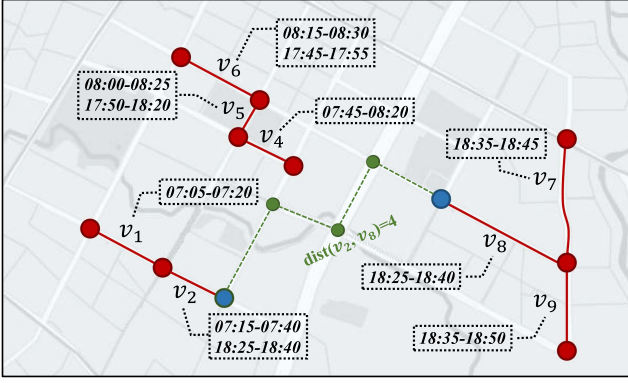


Fig. 2. Spatiotemporal distribution of congestion events during one day. The text in rectangular boxes annotates the start and end times of each congestion event on the road segments.

combinatorial explosion when applied to large-scale urban road networks.

C. Reinforcement Learning

Reinforcement learning has been widely adopted in intelligent transportation systems (ITS) due to its advantages in sequential decision-making and exploration of complex spaces. Its application mainly focuses on traffic signal control (e.g., IntelliLight [35], PressLight [36], and CoLight [13]), order dispatching [37], [38], and travel time estimation [39]. Recently, several studies introduced RL into the field of data mining, using RL agents to conduct effective exploration and path search in knowledge graphs or complex networks [40], [41]. Despite providing methodological inspiration, these RL-based methods cannot be directly transplanted to mining macroscopic congestion propagation patterns in urban road networks.

III. PRELIMINARY

In this section, we describe the key definitions used in this paper and present the problem statement.

Road Segment: A road segment v refers to a section of a road between two adjacent intersections (as shown in Fig. 2). In this study, a two-way road is modeled as two distinct, oppositely directed one-way road segments.

Road Network: The road network is represented as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ is the set of N road segments (vertices), and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents the set of directed spatial connections (edges) between adjacent road segments. The topological structure of $\mathcal{G}$ is described by an adjacency matrix $A \in \{0, 1\}^{N \times N}$, where $A_{i,j} = 1$ if $(v_i, v_j) \in \mathcal{E}$, and $A_{i,j} = 0$ otherwise.

Congestion Event: A congestion event $c = \langle v_i, t^s, t^e \rangle$ characterizes that road segment $v_i \in \mathcal{V}$ experiences traffic congestion starting at time t^s and ending at t^e . Within an observation time window, multiple congestion events may occur on the same road segment. We denote the set of congestion events that occurred on the road segment v_i during the observation period as $\mathbf{c}_i = \{c^1, c^2, \dots\}$. As illustrated in Fig. 2, the congestion

event set of road segment v_1 is $\mathbf{c}_1 = \{\langle v_1, 07:05, 07:20 \rangle\}$. The congestion event set of road segment v_2 can be represented as $\mathbf{c}_2 = \{\langle v_2, 07:15, 07:40 \rangle, \langle v_2, 18:25, 18:40 \rangle\}$.

Congestion Propagation Path: A congestion propagation path reflects the regularity of congestion spreading along a specific direction of urban roadways. We formalize a congestion propagation path as an ordered sequence of road segments, denoted as $\mathcal{P} = \langle v_0, v_1, \dots, v_K \rangle$, where v_k is the segment propagated by congestion at the k -th time step. To ensure physical and topological validity, any inferred path $\mathcal{P}$ must satisfy the following constraints for all $k \in \{1, 2, \dots, K\}$:

- **Directional Consistency:** The propagation must follow the legal traffic direction defined by the directed graph $\mathcal{G}$, i.e., the direction of vehicle travel on road segment $v_k \in \mathcal{P}$ is consistent with its preceding segment v_{k-1} .
- **Spatial Proximity:** The topological distance between consecutive segments in sequence $\mathcal{P}$ must be bounded, formulated as: $1 \leq \text{dist}(v_{k-1}, v_k) \leq \rho$, where $\text{dist}(\cdot, \cdot)$ denotes the shortest path distance between two nodes, and ρ is a predefined spatial distance threshold. As shown in Fig. 2, setting $\rho = 3$ restricts congestion at v_2 to only potentially propagate to v_4 , filtering out unreachable segments like v_8 .

Congestion Propagation Pattern: A congestion propagation pattern is a Directed Acyclic Graph (DAG) that reflects congestion propagation relationships between road segments in the urban road network, which is composed of multiple congestion propagation paths. In this paper, we define it as a directed graph $\mathcal{G}^* = (\mathcal{V}^*, \mathcal{E}^*)$, where $\mathcal{V}^* \subseteq \mathcal{V}$ is the subset of road segments that are actively involved in the congestion propagation process, and $\mathcal{E}^* \subseteq \mathcal{V}^* \times \mathcal{V}^*$ represents the set of inferred virtual links between pairs of road segments in $\mathcal{V}^*$ that exhibit propagation relationships.

Problem Statement: Given the road network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and the historical congestion event log observed over m days, denoted as $\mathbf{C} = \{C_1, C_2, \dots, C_m\}$, we aim to infer the optimal congestion propagation pattern $\mathcal{G}^*$ that best explains the observed spatiotemporal co-occurrence events latent in $\mathbf{C}$,

$$\mathcal{G}^* \leftarrow f(\mathcal{G}, \mathbf{C}) \quad (1)$$

where $C_i \in \mathbf{C}$ represents congestion event data for all segments on the i -th day.

IV. METHOD

A congestion propagation pattern encapsulates multiple overlapping paths to reveal the global, structural laws of congestion spreading. In this section, we model the sequential inference of individual propagation paths as an ordered segment selection process and optimize the decision-making using a reinforcement learning algorithm.

A. Model Formalization

To discover the optimal macroscopic congestion pattern $\mathcal{G}^*$, we formulate the sequential inference of individual congestion propagation paths $\mathcal{P} = \langle v_0, v_1, \dots, v_K \rangle$ as an episodic Markov Decision Process (MDP), which is defined by a 5-tuple $(S, A, \mathcal{T}, R, \gamma)$:

State Space S : The state $s_t \in S$ describes the observable environmental information required for the agent at step t . Since our model focuses on inferring the underlying congestion propagation path, we define the state s_t as the propagation sequence generated before the t -th step: $s_t = (v_0, v_1, \dots, v_t)$.

Action Space A : An action $a_t \in A$ corresponds to selecting the next downstream segment v_{t+1} to append to the propagation path. Crucially, to guarantee that the generated propagation path $\mathcal{P}$ is a simple path without cycles (a fundamental prerequisite for constructing a DAG $\mathcal{G}^*$), we implement a state-dependent propagation constraint rule $\mathcal{F}(s_t)$ (see Section IV-B1 for more details) and define the action $a_t \in A = \mathcal{F}(s_t)$.

State Transition Function $\mathcal{T}$: The state transition function $\mathcal{T} : S \times A \times S \rightarrow [0, 1]$ describes the environment's dynamic. In our problem, the transition is deterministic. Given the current state s_t and an action a_t , the environment transitions to the next state with probability 1, i.e., $\mathcal{T}(s_{t+1}|s_t, a_t) = 1$, where $s_{t+1} = (v_0, v_1, \dots, v_t, v_{t+1})$.

Reward Signal R : The reward function guides the agent to uncover paths that are both historically credible and structurally novel. To align with the macroscopic nature of traffic cascades, we employ a delayed reward mechanism. The agent receives zero immediate reward during traversal; a terminal reward is granted only upon the complete construction of a path $\mathcal{P}$ of length K . The terminal reward comprises two mathematically formalized components:

- 1) Empirical Correlation Reward (R^e). Evaluates the historical fidelity of the path, which is defined as the expected probability of all consecutive transitions on the path. Given the historical congestion event log $\mathbf{C} = \{C_1, C_2, \dots, C_m\}$ over m days, we first select the set of days on which congestion occurs on road u :

$$\mathcal{D}_n(u) = \{i | u \in C_i\}. \quad (2)$$

Then, we define the set of days during which a co-occurrence event occurred within a time-lag τ as:

$$\mathcal{D}_d(u, v | \tau) = \{j \in \mathcal{D}_n(u) | v \in C_j \wedge 0 < t_v^j - t_u^j \leq \tau\}, \quad (3)$$

where t_u^i is the congestion time of road u in the i -th day. Thus, we calculate the empirical propagation probability from segment u to v as:

$$\phi(u, v) = \frac{|\mathcal{D}_d(u, v | \tau)|}{|\mathcal{D}_n(u)|}. \quad (4)$$

Finally, we derive the empirical correlation reward by:

$$R^e(\mathcal{P}) = \frac{1}{K} \sum_{k=1}^K \phi(v_{k-1}, v_k). \quad (5)$$

- 2) Structural Novelty Penalty (R^s). Penalizes redundant exploration. Let $\mathcal{E}^*$ denote the edge set of the current maintained global pattern. We utilize an indicator function $\mathbb{I}_{[\cdot]}$ to penalize the agent proportionally to the ratio of previously discovered edges in $\mathcal{P}$:

$$R^s(\mathcal{P}) = -\frac{1}{K} \sum_{k=1}^K \mathbb{I}_{[(v_{k-1}, v_k) \in \mathcal{E}^*]}. \quad (6)$$

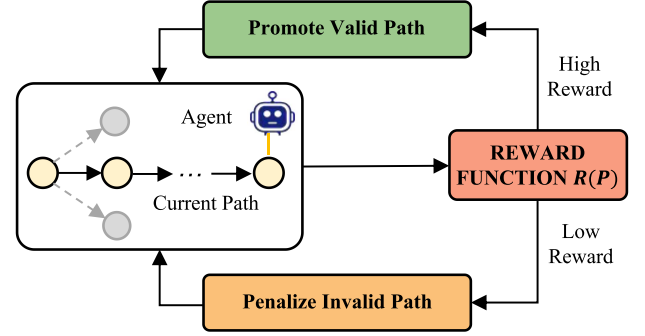


Fig. 3. Reward-guided propagation sequence generation.

$R^e(\mathcal{P})$ provides a delayed positive signal by quantifying the average spatial-temporal transition probability of a completely generated path based on real-world event logs. This signal effectively guides the agent to optimize its action policy, inherently biasing future sequential decisions to prioritize segments that faithfully replicate historically validated congestion cascades. Conversely, $R^s(\mathcal{P})$ actively suppresses topological redundancy. By penalizing the rediscovery of existing edges in $\mathcal{P}$, it forces the agent to shift its probability mass toward unexplored network branches. The comprehensive terminal reward for episode completion is the weighted sum:

$$R(\mathcal{P}) = \begin{cases} \alpha \times R^e(\mathcal{P}) + (1 - \alpha) \times R^s(\mathcal{P}), & \text{if } \mathcal{P} \text{ is terminal} \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where $\alpha \in [0, 1]$ balances historical exploitation and topological exploration. Fig. 3 details the reward-driven optimization mechanism for congestion propagation sequence inference. The agent's policy is iteratively updated via reward feedback, where high rewards upweight valid propagation paths, and low penalties suppress invalid sequences.

Discount Factor γ : The discount factor $\gamma \in [0, 1]$ plays a critical role in balancing instance rewards and future potential returns. The larger the discount factor, the greater the influence of future returns on current decision-making; Conversely, if γ approaches 0, the model will focus only on immediate rewards.

B. Model Framework

Considering the high-dimensional action space arising from large-scale urban road networks, we adopt policy-based reinforcement learning algorithms to update the model through direct policy parameterization and optimization. Fig. 4 displays the overall framework of our proposed method based on the Advantage Actor-Critic (A2C) algorithm.

1) **Action Decision:** The actor network serves as the core component of the action decision-making process, responsible for making the most appropriate decisions based on learned knowledge. Fig. 5 illustrates the structural details of the actor network. As we can see, the actor network integrates environmental observations and road network information to make action decisions.

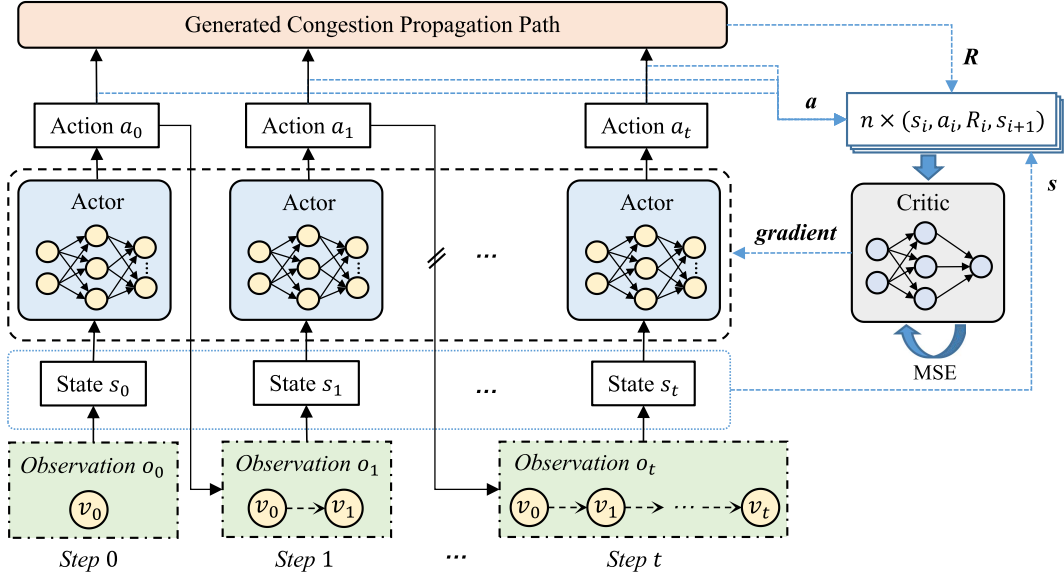


Fig. 4. Framework of the congestion propagation path generation.

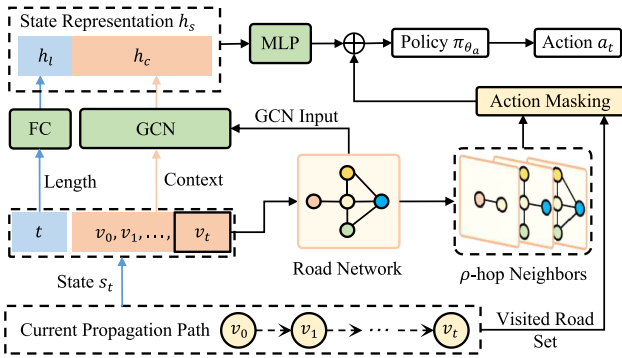


Fig. 5. The structure of the actor network.

At each time step t , the actor network first gets the state information s_t of the current propagation path, including path length information $[t]$ and spatial context information $[v_0, v_1, \dots, v_{t-1}, v_t]$. It then inputs the road network with features into the GCN module to derive feature representations that encapsulate the spatial environment of the roads. Given a road network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and the feature $X \in \mathbb{R}^{N \times D}$ of all roads in $\mathcal{G}$, the process of feature extraction by the GCN can be modeled as follows:

$$\begin{cases} H^{(l+1)} = \sigma(\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)}), \\ \hat{A} = A + I_N, \\ \hat{D}_{ii} = \sum_j \hat{A}_{ij}, \end{cases} \quad (8)$$

where $H^{(l)} \in \mathbb{R}^{N \times D_l}$ is output of the l -th layer of the GCN module and $H^{(0)} = X$, $I_N \in \mathbb{R}^{N \times N}$ is an identity matrix, $\sigma(\cdot)$ denotes activation function (such as ReLU and Sigmoid), $W^{(l)}$ is the learnable parameter of the l -th GCN layer.

The actor network then utilizes the spatial context information $\mathbf{v}_c = [v_0, v_1, \dots, v_{t-1}, v_t]$ to retrieve the corresponding spatial

latent state representation h_c from the output features of the GCN module. Simultaneously, it feeds the path length information $\mathbf{v}_l = [t]$ into a fully connected layer (FC) to obtain the length latent state representation h_l . Then we concatenate h_c and h_l to form a unified state representation h_s . Finally, a Multi-Layer Perceptron (MLP) with multiple nonlinear activation functions takes h_s as input and outputs the raw action logits $z_t \in \mathbb{R}^N$ over the entire road network:

$$h_c = \text{MEAN}(H(\mathbf{v}_c, :)), \quad (9)$$

$$h_l = \sigma(W_l(\mathbf{v}_l) + b_l), \quad (10)$$

$$h_s = [h_c \parallel h_l], \quad (11)$$

$$z_t = \text{MLP}(h_s; \theta), \quad (12)$$

where σ represents the non-linear activation function (here we use the ReLU function), W_l and b_l are the learnable parameters of the fully connected layer for embedding path length context information, $\parallel$ denotes the concatenation operation, θ denotes the learnable parameters of the MLP.

To enforce strict physical topological constraints and prevent cyclic traversals, we design a propagation constraint rule $\mathcal{F}(s_t)$. This rule determines the valid action space of the agent at step t . Specifically, given the state $s_t = (v_0, v_1, \dots, v_{t-1}, v_t)$ and a predefined spatial distance threshold ρ , we construct the propagation constraint rule as follows:

$$\mathcal{F}(s_t) = \{v \in \mathcal{V} | \text{dist}(v_t, v, \text{dir}) \leq \rho \wedge v \notin s_t\}, \quad (13)$$

where $\text{dist}(v_t, v, \text{dir})$ gives the shortest path distance between v_t and v along the direction dir (upstream/downstream).

Based on the propagation constraint rule $\mathcal{F}(s_t)$, we update the action policy $\pi_{\theta_a}(\mathbf{a} | s_t)$ via a masking mechanism, ensuring that invalid nodes have exactly zero probability of being selected:

$$\pi_{\theta_a}(\mathbf{a} | s_t) = \text{softmax}(\text{Masking}(z_t, \mathcal{F}(s_t))), \quad (14)$$

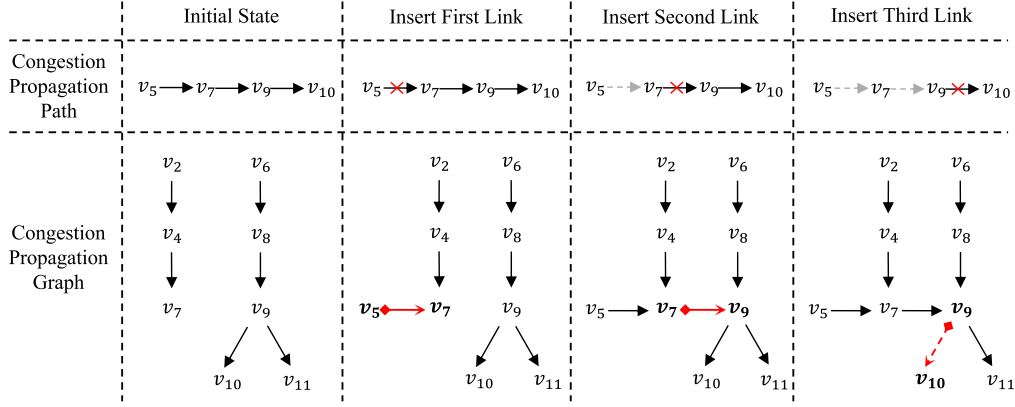


Fig. 6. Illustration of congestion propagation path aggregation. The red arrow is the newly inserted link in the congestion propagation pattern graph.

Algorithm 1: Action Selection.

Input: Road network $\mathcal{G}$, Node embeddings H , Current propagation path $\mathcal{P}$, Current state s_t
Output: Selected action a_t

- 1: Obtain h_c via (9) using H
- 2: Get length embedding h_l according to (10)
- 3: Construct state representation vector h_s via (11)
- 4: Calculate raw logits z_t by a MLP according to (12)
- 5: Determine the propagation constraint rule $\mathcal{F}(s_t)$ based on the current state s_t via (13)
- 6: Apply rule masking and calculate action probabilities $\pi_{\theta_a}(a|s_t)$ according to (14)
- 7: Sample action $a_t \sim \pi_{\theta_a}(\cdot|s_t)$
- 8: **return** a_t

where Masking function penalizes invalid actions with negative infinity and softmax function is applied to normalize the masked logits.

According to the updated action policy $\pi_{\theta_a}(a|s_t)$, the agent samples an action a_t as the next downstream segment to extend the congestion propagation path. In our problem, the path generation episode terminates when either of the following criteria is met:

- *Topological Dead-end:* The valid action space becomes empty (i.e., $\mathcal{F}(s_t) = \emptyset$), indicating no reachable unvisited downstream neighbors exist.
- *Statistical Insignificance:* The latest propagation pair (v_{t-1}, v_t) exhibits weak historical correlation. Specifically, we evaluate the co-occurrence confidence between v_{t-1} and v_t using the Jaccard similarity of their historical congestion event sets. Let $\mathbf{C}^{v_{t-1}}$ and $\mathbf{C}^{v_t}$ be the historical congestion event sets of two road segments, respectively. The episode truncates if:

$$Jaccard(v_{t-1}, v_t) = \frac{|\mathbf{C}^{v_{t-1}} \cap \mathbf{C}^{v_t}|}{|\mathbf{C}^{v_{t-1}} \cup \mathbf{C}^{v_t}|} < \epsilon, \quad (15)$$

where ϵ is a predefined statistical significance threshold (we choose $\epsilon = 0.5$ here).

Algorithm 1 details the action selection process of the actor network. The actor network generates state embeddings in lines 1-3. Line 4 describes the process of generating raw logits. Line 5 constructs the propagation constraint rule. Lines 6-7 illustrate how the actor network selects and executes an action based on the policy, and then obtains a reward and transitions to the next state.

As formulated in Section IV-A, the macroscopic pattern $\mathcal{G}^*$ is dynamically constructed by aggregating individual inferred paths. Let $\mathcal{G}_i^* = (\mathcal{V}_i^*, \mathcal{E}_i^*)$ denote the global propagation pattern maintained after the i -th episode. Upon the successful generation of a high-confidence path $\mathcal{P}$ comprised of edge set $\mathcal{E}_{\mathcal{P}}$, the global pattern is updated via a union operation: $\mathcal{E}_{i+1}^* = \mathcal{E}_i^* \cup \mathcal{E}_{\mathcal{P}}$. During this integration, redundant edges already present in $\mathcal{E}_i^*$ are inherently ignored (e.g., $v_9 \rightarrow v_{10}$ in Fig. 6), while novel directed edges successfully bridge isolated subgraphs or expand the cascaded tree (e.g., $v_7 \rightarrow v_9$ in Fig. 6), thereby iteratively enriching the topological completeness of the global congestion pattern. Furthermore, we perform cycle detection as each cascading link is integrated into graph $\mathcal{G}_i^*$. If the insertion of a link introduces a directed cycle, the link is discarded, and the merging process for the remainder of the current path is immediately aborted.

2) *Model Optimization:* The critic network is the core component of the model optimization, which functions to evaluate the actor's policy according to interaction data collected by the agent and calculates loss to update the network parameters for the actor and the critic network. Given the interaction records spanning n consecutive timesteps, the critic network first evaluates the value $V_{\theta_c}(s_i)$ of the state s_i and the $V_{\theta_c}(s_{i+n})$ of the terminal state s_{i+n} after n -steps interactions. Subsequently, the Critic derives the target value y_i for the agent in the initial state s_i by integrating immediate rewards obtained during the n -step interactions and $V_{\theta_c}(s_{i+n})$:

$$y_i = \sum_{j=1}^n \gamma^{(j-1)} R_{i+j} + \gamma^{(n)} V_{\theta_c}(s_{i+n}), \quad (16)$$

where θ_c denotes the parameters of the critic network. Based on the estimation value $V_{\theta_c}(s_i)$ computed by the critic network and

the target value y_i , the agent calculates the temporal difference (TD) error as follows:

$$\delta_i = y_i - V_{\theta_c}(s_i). \quad (17)$$

A larger δ_i indicates that the critic's evaluation of the current state value is more inaccurate. Therefore, we optimize the parameters θ_c of the critic network by minimizing the Mean Squared Error (MSE) loss:

$$L(\theta_c) = \frac{1}{2}(\delta_i)^2. \quad (18)$$

In the congestion propagation path inference problem, the actor network aims to find an optimal policy to guide the agent in generating the propagation path that can best explain the congestion events in observation data. As described in IV-A, this objective is formally defined as:

$$J(\theta_a) = \mathbb{E}_{s_t, a_t \sim \pi_{\theta_a}} \left[\sum_{t=0}^T R(s_t, a_t) \right], \quad (19)$$

where π_{θ_a} is the policy used for guiding the action decision of the agent, and we denote its parameters as θ_a . a and s represent the action selected by the policy and the resulting state after action execution, respectively. T specifies the total steps for completing a propagation path construction process. The gradient of the objective function is derived as:

$$\nabla J(\theta_a) = \mathbb{E}_{s_t, a_t \sim \pi_{\theta_a}} \left[\sum_{t=0}^T A^{\pi_{\theta_a}}(s_t, a_t) \nabla_{\theta_a} \log \pi_{\theta_a}(a_t | s_t) \right], \quad (20)$$

where $A^{\pi_{\theta_a}}(s_t, a_t)$ represents the advantage function, which measures the advantage of taking action a_t over other actions at state s_t . In this study, we replace the advantage value with the TD error to evaluate the policy of the actor network. Additionally, we employ action masking and probability renormalization techniques to update the policy. Therefore, we have:

$$\nabla J(\theta_a) = \mathbb{E}_{s_t, a_t \sim \pi_{\theta_a}} \left[\sum_{t=0}^T \delta_t \nabla_{\theta_a} \log \pi_{\theta_a}(a_t | s_t) \right]. \quad (21)$$

To maximize the cumulative reward, we use gradient ascent to increase the probability of actions with high advantages.

V. EXPERIMENTS

In this section, we comprehensively evaluate the proposed approach using both real-world and synthetic datasets. Our codes and data are available online.¹

A. Datasets

Real-world Datasets: We employ two GPS trajectory datasets from Xi'an and Chengdu, which contain complex empirical traffic flow data with inherent noise. We estimate the historical travel speed on each road at 5-minute intervals and extract the congestion timetable. Due to regular commuting behaviors, urban traffic congestion is heavily concentrated during the

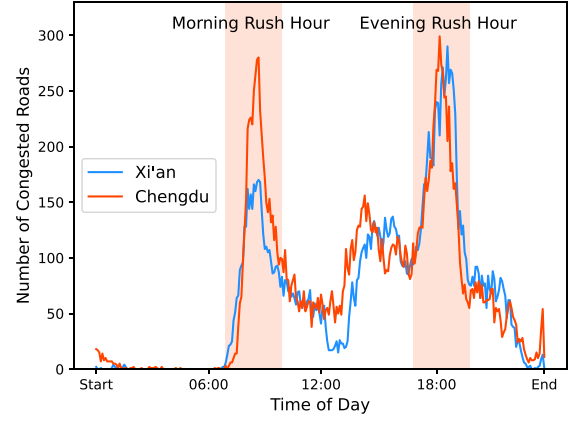


Fig. 7. Temporal distribution of congested road segments over one day.

morning (7:00–10:00) and evening (17:00–20:00) peak hours, as illustrated in Fig. 7. Consequently, we specifically evaluate the performance of our proposed method during these critical periods to demonstrate its practical utility. For all experiments, the dataset is chronologically partitioned, with 75% allocated for model training and the remaining 25% reserved for evaluation.

Simulated Datasets: To establish a rigorous ground truth for congestion propagation pattern inference, we employ two classic directed network models—the Forest Fire Graph model and the Kronecker Graph model—to synthesize cascading graphs. For each model, we generate a base network comprising 500 nodes and 500 edges, assigning each edge a random propagation probability ranging from 0.1 to 0.9. Furthermore, to emulate the inherent noise of real-world traffic environments, we inject 250 random spurious edges into each base graph, assigning them marginal propagation probabilities ($p < 0.1$). Finally, we simulate a series of cascading events across these synthetic networks, recording the exact activation timestamp for each edge.

B. Baselines and Variants

To evaluate the performance of our proposed method, we conducted comparative experiments using currently recognized best-performing baseline models.

- **STC_DBN [17]:** This approach first constructs spatiotemporal congestion trees (STCTrees) leveraging the spatiotemporal continuity of congestion, then applies an Apriori-based frequent pattern mining algorithm to identify recurrent substructures within STCTrees.
- **CasInf [18]:** This method detects congestion propagation patterns through submodular optimization. It first calculates the likelihood of congestion transmission between road segments under spatiotemporal constraints, then applies a submodular optimization technology to select k casual links with the highest marginal gain.
- **SPPTC [33]:** This approach first extracts the temporal duration and correlation strength of congestion events on road segments, then detects propagation relationships between congestion events using the Dynamic Time Warping

¹https://github.com/QLtan/GRL_CPP

(DTW) algorithm, and finally constructs a spatiotemporal congestion propagation graph.

- *CAW-N [3]*: This method evaluates the congestion correlation between roads by applying spatial thresholds and a specific congestion propagation speed threshold (the ratio of spatial distance to the time delay in congestion onset). Directed links are then established between segments that meet these spatio-temporal criteria. Finally, all identified propagation relationships are globally aggregated, yielding a congestion propagation graph.

In GRL_CPP, we developed several modules to enhance the model efficiency and optimize congestion propagation pattern analysis. To validate the effectiveness of these modules, we constructed several variants and conducted comprehensive evaluations on these variants within the dataset.

- *GRL_CPP w/o GCN*: This variant replaces the GCN module with a fully connected neural network for state feature representation, while keeping the remaining components of the model unchanged.
- *GRL_CPP w/o Mask*: This variant removes the action masking module in the actor network and imposes penalties via reward signals on congestion propagation instances exceeding spatial boundaries, thereby guiding the agent to explore within appropriate spatial ranges.

C. Evaluation Metrics

Metrics for Real-world Datasets: To evaluate the performance of all methods on real-world datasets, we improve the $score@k$ metric proposed in [18], enabling its application to recurrent congestion scenarios. Given a congestion propagation graph $\mathcal{G}^*$ containing k edges and historical congestion data $\mathbf{C}$, $score@k$ quantifies the average occurrence probability of the k congestion propagation path pairs in $\mathcal{G}^*$ within the historical congestion information $\mathbf{C}$.

$$score@k(\mathcal{G}^*) = \frac{\sum_{(i,j) \in \mathcal{G}^*} prob_{i,j}}{k}, \quad (22)$$

where $prob_{i,j}$ denotes the occurring probability of the correlation link $v_i \rightarrow v_j$ over m days and can be calculated by:

$$prob_{i,j} = \frac{\#prop_{i,j}(\tau)}{\#count(i, \bar{j})}, \quad (23)$$

where $\#count(i, \bar{j})$ quantifies the number of occurrences in historical congestion data where road segment v_i was congested while v_j remained smooth before the congestion event on v_i . $\#prop_{i,j}(\tau)$ denotes the total number of times in historical data that v_j also became congested within a τ time interval following the onset of congestion on v_i .

In addition, we provide the average occurrence frequency of all identified propagation links, serving as an indicator of the statistical significance of these propagation relationships.

$$Frequency = \frac{\sum_{(i,j) \in \mathcal{G}^*} \#prop_{i,j}(\tau)}{k \times |\mathbf{C}|}. \quad (24)$$

Metrics for Simulated Datasets: To evaluate whether the model correctly identifies the true congestion propagation edges,

we treat it as a graph recovery task. We report the F1-score by comparing the inferred propagation graph against the simulated ground-truth, which can be calculated as follows.

$$\begin{aligned} \text{Precision} &= \frac{N_{TP}}{N_{TP} + N_{FP}}, \quad \text{Recall} = \frac{N_{TP}}{N_{TP} + N_{FN}}, \\ F1\text{-score} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \end{aligned} \quad (25)$$

where N_{TP} refers to the number of edges that are correctly inferred by the algorithm, N_{FP} is the number of the wrong inferred edges that are not in the real propagation graph, and N_{FN} denotes the true edges that are not correctly inferred by the algorithm. A higher F1-score indicates more accurate mechanism extraction.

D. Overall Performance

1) Results on Real-World Data: Driven by the underlying physical road topology and residents' travel demands, traffic congestion exhibits complex spatial directionality. Consequently, our experiments explicitly account for both the forward (downstream) and backward (upstream) propagation dynamics of congestion. Furthermore, to evaluate how the recurrence rate of congestion influences the extraction of propagation patterns, we benchmark all methods across two distinct scenarios:

- *Scenario 1 (Recurrent Congestion)*: A road segment can experience multiple congestion events within the observation period.
- *Scenario 2 (Single-Instance Congestion)*: A road segment experiences only a single congestion event within the observation period.

Forward Propagation Analysis: Table I summarizes the overall performance of different methods in analyzing forward congestion propagation patterns under two scenarios, with the best results highlighted in bold and the suboptimal results underlined. The values in the brackets represent the average occurrence frequency of each propagation link in the constructed cascading propagation graph within the test dataset.

The top half of the table presents the performance of all comparison methods in Scenario 1 across different datasets. The results show that GRL_CPP consistently outperforms all baselines by a significant margin. Specifically, GRL_CPP achieves remarkable average performance improvements of 42.60%, 102.42%, and 84.62% over STC_DBN, CasInf, and SPPTC, respectively. Even when compared to the strongest baseline, CAW-N, GRL_CPP maintains a substantial lead (0.8773 vs. 0.6653 on average). This demonstrates the critical necessity of modeling sequential congestion propagation impacts and effectively capturing the holistic congestion context, rather than treating propagation relationships as independent events.

The bottom half of the table details the experimental results in Scenario 2. As expected, all methods experience a noticeable performance degradation due to the drastically reduced number of historical congestion events. Despite this data scarcity, GRL_CPP consistently maintains optimal performance across all datasets (averaging 0.6007), underscoring the exceptional robustness and generalization capability of our approach. Among

TABLE I
FORWARD CONGESTION PROPAGATION PATTERN ANALYSIS ACROSS DIFFERENT METHODS

Scenarios	Methods	Xi'an		Chengdu		Average
		Morning Peak	Evening Peak	Morning Peak	Evening Peak	
Scenario 1	STC_DBN	0.5610 (2.13)	0.6287 (3.37)	0.6328 (2.66)	0.6381 (3.25)	0.6152 (2.85)
	CasInf	0.4314 (0.97)	0.4234 (1.21)	0.4373 (1.08)	0.4414 (1.28)	0.4334 (1.14)
	SPPTC	0.4040 (1.43)	0.4775 (2.47)	0.4940 (1.99)	0.5253 (2.86)	0.4752 (2.19)
	CAW-N	<u>0.6204</u> (2.74)	<u>0.6799</u> (4.06)	<u>0.6749</u> (3.23)	<u>0.6860</u> (4.17)	<u>0.6653</u> (3.55)
	GRL_CPP	0.8546 (1.43)	0.8868 (1.92)	0.8692 (1.60)	0.8987 (1.76)	0.8773 (1.68)
Scenario 2	STC_DBN	0.3146 (0.26)	<u>0.3854</u> (0.37)	0.3695 (0.34)	0.3374 (0.32)	0.3517 (0.32)
	CasInf	0.3406 (0.30)	0.3644 (0.34)	0.3808 (0.34)	0.3521 (0.33)	0.3595 (0.33)
	SPPTC	0.3303 (0.31)	0.3770 (0.37)	0.4457 (0.44)	<u>0.3915</u> (0.39)	0.3861 (0.38)
	CAW-N	<u>0.4039</u> (0.39)	0.3439 (0.34)	<u>0.5326</u> (0.53)	0.2977 (0.30)	<u>0.3945</u> (0.39)
	GRL_CPP	0.6146 (0.46)	0.5974 (0.50)	0.6312 (0.50)	0.5596 (0.46)	0.6007 (0.48)

TABLE II
BACKWARD CONGESTION PROPAGATION PATTERN ANALYSIS ACROSS DIFFERENT METHODS

Scenarios	Methods	Xi'an		Chengdu		Average
		Morning Peak	Evening Peak	Morning Peak	Evening Peak	
Scenario 1	STC_DBN	0.5509 (2.08)	0.6325 (3.33)	0.6331 (2.55)	0.6235 (3.26)	0.6100 (2.81)
	CasInf	0.3573 (0.90)	0.3681 (1.28)	0.3504 (0.97)	0.3425 (1.19)	0.3546 (1.09)
	SPPTC	0.4400 (1.48)	0.5273 (2.67)	0.5294 (2.08)	0.5651 (2.99)	0.5155 (2.31)
	CAW-N	<u>0.6071</u> (2.68)	<u>0.6682</u> (3.95)	<u>0.6647</u> (3.16)	<u>0.6810</u> (4.18)	<u>0.6553</u> (3.49)
	GRL_CPP	0.8252 (1.52)	0.8762 (1.86)	0.8535 (1.59)	0.8717 (1.78)	0.8567 (1.69)
Scenario 2	STC_DBN	0.3014 (0.27)	0.3654 (0.35)	0.4178 (0.39)	0.3345 (0.32)	0.3548 (0.33)
	CasInf	0.3281 (0.29)	<u>0.3950</u> (0.38)	0.3890 (0.37)	0.3509 (0.33)	0.3658 (0.34)
	SPPTC	0.3373 (0.31)	0.3772 (0.37)	0.4553 (0.45)	<u>0.3783</u> (0.38)	<u>0.3870</u> (0.38)
	CAW-N	<u>0.3718</u> (0.36)	0.3474 (0.34)	<u>0.5265</u> (0.52)	0.2922 (0.29)	0.3845 (0.38)
	GRL_CPP	0.6015 (0.46)	0.5814 (0.49)	0.6114 (0.50)	0.5495 (0.45)	0.5860 (0.48)

the baselines, an interesting shift occurs: SPPTC, CAW-N, and CasInf generally outperform STC_DBN in this sparse scenario. This shift happens because sparse congestion events severely restrict the scale and completeness of the causal congestion trees constructed by STC_DBN, thereby degrading the accuracy of its local propagation extraction. Under these highly constrained conditions, models like SPPTC and CasInf, which explicitly model complex spatio-temporal constraints, exhibit slightly better resilience than STC_DBN.

Backward Propagation Analysis: Table II presents the performance of all methods in exploring the backward congestion propagation patterns across two scenarios. Overall, the performance distribution across all baselines closely mirrors that of the forward propagation analysis (Table I). This consistency confirms the nature of urban traffic dynamics: congestion shockwaves not only spread forward along the traffic flow but also frequently spill backward to upstream road segments.

In Scenario 1 (recurrent congestion), GRL_CPP achieves an average score@500 of 0.8567, significantly outperforming the strongest baseline, CAW-N (0.6553). An insightful conclusion emerges when analyzing the occurrence frequency (in parentheses): while CAW-N and STC_DBN exhibit high average

frequencies (3.49 and 2.81, respectively), their overall performance remains suboptimal. This indicates that traditional methods tend to extract redundant and spurious correlation links, confusing mere co-occurrence with actual propagation. In stark contrast, GRL_CPP yields a lower, refined frequency (1.69) alongside the highest performance. This demonstrates our method's exceptional capability to effectively filter out structural noise and precisely isolate the true causal propagation rules amidst complex traffic environments.

Furthermore, in Scenario 2 (non-recurrent congestion scenario), where event information is highly insufficient, all baseline models suffer substantial performance degradation. Remarkably, GRL_CPP not only sustains the optimal evaluation score (0.5860 on average) but also achieves the highest propagation significance frequency (0.48) among all models. This contrast perfectly highlights the robustness of our inductive design: when data is abundant (Scenario 1), GRL_CPP acts as a strict filter to eliminate noise; when data is extremely scarce (Scenario 2), it leverages its learned causal mechanisms to actively discover hidden structural dependencies. The results prove that our method provides highly robust, bidirectional congestion pattern analysis regardless of data sparsity.

TABLE III
RECONSTRUCTION PERFORMANCE (F1-SCORE) ON SIMULATED CASCADING
GRAPHS AMONG VARIOUS APPROACHES

Methods	F1-score		Average
	Kronecker	Forest Fire	
STC_DBN	0.4612	0.3660	0.4136
CasInf	0.2710	0.2605	0.2658
SPPTC	0.2200	0.2056	0.2128
CAW-N	0.4520	0.3160	0.3840
GRL_CPP	0.7140	0.7260	0.7200

2) *Validation of Propagation Mechanism on Simulated Data:* We evaluate the models' ability to reconstruct the actual congestion propagation graph using two simulated datasets described in Section V-A. The structural recovery results, evaluated using F1-score, are presented in Table III. It is crucial to note that in this experiment, the inferred graphs are strictly constrained to have the exact same number of edges as the ground truth; consequently, Precision, Recall, and the F1-score are mathematically equivalent. As demonstrated, our approach achieves the highest average F1-score of 0.7200 in reconstructing the true congestion propagation graphs. This represents a massive 74.1% relative improvement over the best-performing baseline, STC_DBN (average F1-score: 0.4136). Moreover, we found that all existing baselines (STC_DBN, CasInf, SPPTC, and CAW-N) score below 0.5 across both simulated datasets, which indicates that more than half of the propagation edges inferred by these methods are entirely incorrect (spurious). Models like CasInf and SPPTC perform particularly poorly (averaging around 0.2), proving that traditional spatio-temporal models heavily conflate mere statistical co-occurrence with actual sequential propagation. They mistakenly link road segments that simply happen to be congested simultaneously, failing to capture the true causal cascading mechanisms.

E. Ablation Study

Fig. 8 displays the training curves of GRL_CPP, GRL_CPP w/o GCN (model without GCN), and GRL_CPP w/o Mask (model without action masking) under two scenarios using datasets from Xi'an and Chengdu, with 500 episodes for each run. The results demonstrate that the model lacking the GCN module exhibits more oscillatory training trajectories and significantly slower convergence across all scenarios. This conclusively highlights the critical role of the GCN module in stabilizing training and enhancing model performance. This is attributed to GCN's powerful graph-based feature extraction capability, which enables the model to comprehend complex environments and improve decision-making.

Furthermore, the GRL_CPP w/o Mask model, deprived of the action masking module, struggles to identify optimal policies for environmental exploration within complex action spaces, as evidenced by its persistently low reward values throughout training. This indicates that the action masking module effectively constrains action feasibility and substantially boosts the agent's

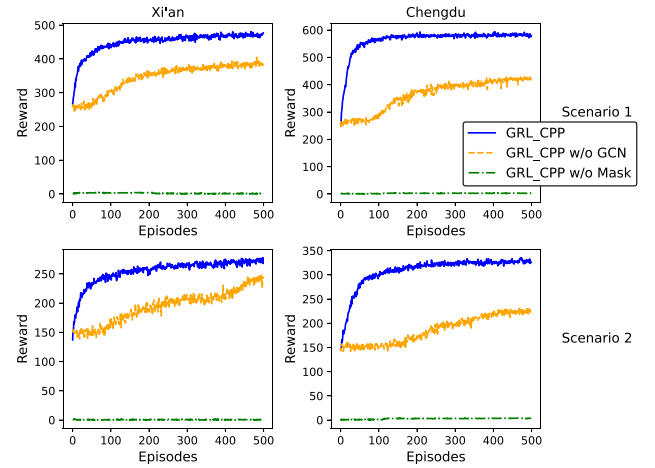


Fig. 8. Training performance of GRL_CPP and its variants under different datasets and scenarios.

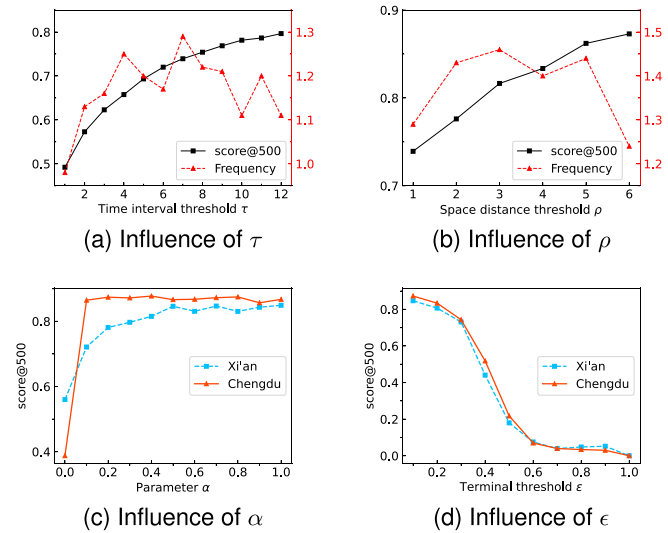


Fig. 9. Studies on the influence of model parameters.

exploration efficiency. Collectively, both the GCN module and the action masking module are indispensable components of GRL_CPP, each exerting a significant impact on the method's overall performance.

F. Parameter Analysis

In this section, we study the impact of model parameters on performance, focusing on the following key components:

1) *Different Time Interval Threshold τ :* We analyze the impacts of different temporal interval thresholds τ on $score@500$ and Frequency. Fig. 9(a) shows the results obtained by GRL_CPP in the Xi'an dataset when analyzing forward congestion propagation patterns under varying τ values. As τ increases, the $score@500$ value improves, while Frequency first rises and then declines. This occurs because τ controls the temporal window size for congestion propagation: a larger τ allows longer propagation processes between road segments, increasing

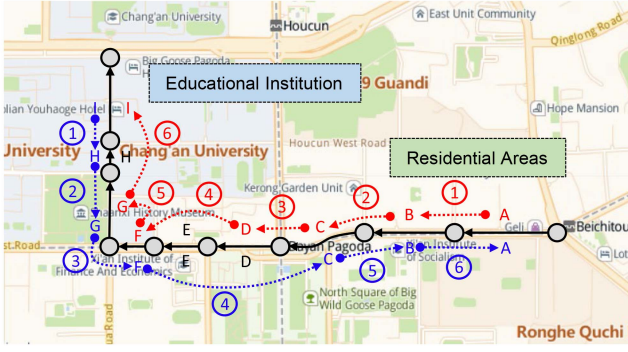


Fig. 10. An instance of congestion propagation paths found in the morning rush hour, where black arrows indicate the roads with directions, red/blue arrows represent forward/backward congestion propagation links. The number in the circle specifies the sequence of congestion propagation.

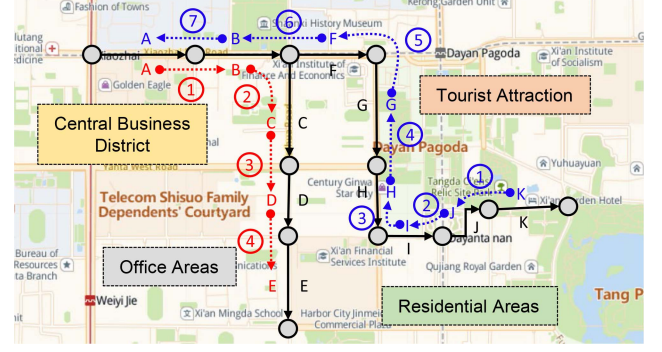


Fig. 11. Congestion propagation paths found in the evening rush hour.

the likelihood of congestion spread. However, an excessively large τ introduces unrealistic propagation links, reducing the significance of identified congestion patterns. To obtain highly relevant congestion propagation patterns while maintaining their significance, we primarily select $\tau = 7$ (35 minutes) for our experiments.

2) *Different Space Distance Threshold ρ* : Fig. 9(b) illustrates how model performance varies with the parameter ρ . Increasing ρ relaxes the spatial constraints for congestion propagation, enabling congested roads to directly propagate their influence to ρ -hop neighbors. However, excessively large ρ results in unrealistic propagation relationships that cannot accurately represent real-world congestion phenomena. This explains why *score@500* gradually increases while Frequency initially increases before decreasing. Consequently, we choose $\rho = 3$ for experiments to ensure both the relevance and significance of the generated congestion propagation patterns.

3) *Different Weight Factor α* : Fig. 9(c) shows the impact of reward weight factors α on model performance based on experiments conducted on Xi'an and Chengdu datasets. A smaller α allows the model to focus on discovering novel congestion propagation relationships, while excessively large α restricts the model to reinforcing previously explored high-correlation congestion propagation patterns. This aligns with the observed trend where model performance first improves and then slightly fluctuates as α increases. Consequently, we set $\alpha = 0.7$ in our experiments.

4) *Different Terminal Threshold ϵ* : Fig. 9(d) reveals that as ϵ increases, *score@500* exhibits a sharp decline. This occurs because ϵ defines the minimum correlation threshold that each congestion propagation link within the propagation path must satisfy. A larger ϵ imposes stricter criteria for establishing congestion propagation relationships, resulting in fewer qualified links that satisfy these conditions.

G. Case Studies

In this section, we elaborate on the discovered propagation patterns and explain why congestion propagation in this area

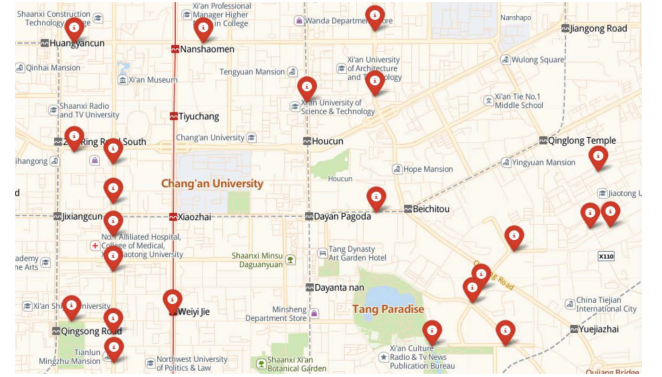


Fig. 12. Spatial mapping of key roads in the forward propagation graph.

follows such patterns. Fig. 10 illustrates forward (red) and backward (blue) propagation paths during Xi'an's morning peak, revealing underlying commuting dynamics. The forward cascade ($A \rightarrow B \rightarrow \dots \rightarrow I$) spreads from residential to educational areas, fundamentally driven by converging morning commuter traffic. In contrast, the backward sequence ($I \rightarrow H \rightarrow \dots \rightarrow A$) reflects an upstream propagation against the traffic flow. This is primarily caused by downstream blockages that trigger vehicle accumulation and subsequent queue spillbacks.

Fig. 11 illustrates evening peak congestion spreading from commercial to residential zones due to homeward commutes, matching the forward sequence ($A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$). Furthermore, the convergence of these commuters with heavy night-time tourist traffic near a local landmark creates severe bottlenecks. This compounded volume drives the backward propagation path from K to A via upstream queue spillbacks.

To validate the physical significance of our model, we mapped the top 5% of roads most frequently involved in morning forward propagation paths onto Xi'an's real-world network (Fig. 12). Spatially, these key nodes cluster around dense residential, educational, and office zones, acting as critical bottlenecks driven by massive commuting flows. Notably, these propagation nodes are predominantly vulnerable secondary roads traversing dense neighborhoods rather than wide arterials. Due to their limited capacity, these secondary roads quickly oversaturate under heavy commuting demand, triggering outward congestion cascades. This discovery provides actionable insights for urban

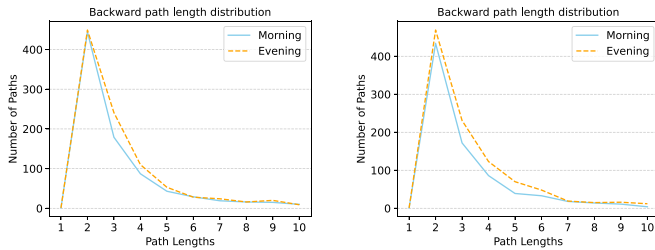


Fig. 13. Distribution of path lengths in forward/backward directions.

management: targeted optimization of this 5% of “key roads” can effectively cut off the propagation chain at the lowest cost and alleviate network-wide congestion.

We also conducted experiments on the Xi’an dataset to analyze the path length distribution and display the results in Fig. 13. From this figure, we can find that although both the morning peak and the evening peak exhibit a high frequency of short, localized bottlenecks (at length 2), the evening peak consistently demonstrates a “fatter tail”—meaning it generates a significantly higher volume of long propagation paths (lengths ≥ 3) in both directions. This phenomenon perfectly captures human travel behavior: morning commutes are typically highly concentrated and point-to-point (from home to specific employment hubs), causing severe but localized jams. In contrast, evening commutes are more dispersed and heterogeneous (involving trips to commercial areas, dining, and diverse residential zones), which creates complex, cascading bottlenecks that propagate much further across the urban network. Moreover, the remarkable structural similarity between the forward (downstream cascading) and backward (upstream queue spillback) distributions underscores that congestion spreads against the traffic flow just as severely as it does along it.

VI. CONCLUSION

This study investigates the problem of uncovering underlying congestion propagation patterns in urban road networks. We conceptualize congestion propagation patterns as combinations of multiple congestion propagation paths, and formulate the inference of each path as a Markov Decision Process to capture long-term spatiotemporal dependencies. Subsequently, we introduce graph convolutional networks to extract heterogeneous environmental features of road segments and develop a novel graph reinforcement learning framework to optimize the decision process. Leveraging the powerful representational capacity of GCNs and the robust exploration capabilities of reinforcement learning, our proposed method successfully identifies propagation patterns that best explain observed traffic congestion phenomena in historical traffic data. Extensive experiments conducted on two real-world datasets demonstrate that GRL_CPP significantly outperforms state-of-the-art baselines, achieving up to a 52% performance gain in bidirectional (forward and backward) propagation pattern mining. Furthermore, evaluations on simulated cascading graphs confirm that GRL_CPP

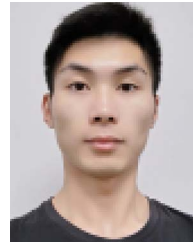
successfully disentangles authentic physical propagation mechanisms from environmental noise by achieving a remarkable F1-score of 0.7200 in true causal graph reconstruction.

In the future, we will focus on two primary directions. First, we aim to develop dynamic congestion cascading pattern mining technology, enabling the real-time updating of congestion propagation graphs to capture sudden traffic anomalies. Second, integrating multi-source exogenous data—such as weather conditions, holidays, and public transit dynamics—into the reinforcement learning environment could further enhance the comprehensiveness and accuracy of urban bottleneck prediction.

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